POLYTREND

PFAS-free PPA TTK-3700E

PERFORMANCE DATA SHEET

SILICONE-BASED PPA

PFAS-FREE

ACTIVE CONTENT 5%

BASIC INFORMATION

PRODUCT CODE	CHEMISTRY	RECOMMENDED DOSAGE	ACTIVE CONTENT
TTK-3700E	Silicone-based	0.5% – 2.0%	5.00%

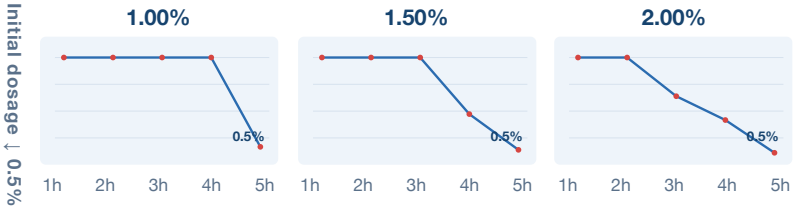
TEST CONDITIONS

FILM SPECIFICATION	PROCESS TYPE	TEST TEMPERATURE	RELATIVE HUMIDITY
IPE 30 μm	Blown film	25 °C	50.00%

PERFORMANCE COMPARISON

Category	Test Item	Untreated Film		TTK-3700E (1%)		Standard
		MD	TD	MD	TD	
Optical	Transmittance (%)	90.30		90.50		GB/T 2410-08
	Haze (%)	5.08		5.06		
Mechanical	Tensile Strength (MPa)	33.21	17.69	32.76	17.74	GB/T 1040.3
	Elongation at Break (%)	111	595	108	599	
	Tensile Modulus (MPa)	321	433	319	429	
	Impact Break Mass (g)	67		66		GB 9639-88
Print-ability	Heat Seal Test (°C, N)	F0.69 · F3.81 · F8.98 10.80 · 12.58 · 12.69		F0.71 · F3.43 · F9.28 10.49 · 12.36 · 11.61		QB/T 2358-98
	CoF — film / film static / kinetic	0.15 / 0.18		0.16 / 0.17		GB 10006-88
	CoF — film / steel static / kinetic	0.32 / 0.34		0.30 / 0.31		
Corona Decay	Dyne Level (mN/m)	Off-line 45 · 3 d 43 · 7 d 42 · 15 d 40 · 21 d 40 · 30 d 40				Dyne pen

INITIAL DOSAGE VS. EFFECT ONSET TIME



Note: Effect onset time is affected by equipment condition, line speed, temperature and other factors. Adjust the dosage in a timely manner based on the actual film surface performance.

USAGE RECOMMENDATIONS

1. Purge the extruder thoroughly with a purging compound before use, and make sure the **die head is completely clean**.
2. Where **printing or heat-sealing** performance is required, start with an initial dosage of **2%** and run for **no less than 2 hours**, then fine-tune the dosage according to the film surface condition.